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MCA (First Semester)

EXAMINATION, Dec. 2023

Paper-MCA-103

(Programming and Problem Solving using
'C' & C++)

Time : Three Hours

Maximum Marks : 60

Minimum Pass Marks : 21

Note— Attempt *All* questions. Question No. 1 is compulsory. Question 2 to 6 carry equal marks.

1. Attempt any *five* questions in brief. $5 \times 2 = 10$
 - (i) What is C++ ?
 - (ii) What is pointer ?
 - (iii) Describe variable in C.

P.T.O.

(2)

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- (iv) What do you mean by data type ?
- (v) Define arrays.
- (vi) What is control statement ?
- (vii) What do you mean by constructor ?
- (viii) Describe arithmetic operators.
- (ix) What do you understand by Bottom up design ?
- (x) What is the benefit of Loop ?

Unit-I

2. (a) Discuss about testing and debugging.
(b) Write an algorithm for calculating and printing of factorial of 5. 10

Or

- (a) Explain about characteristics of a good program.
(b) Discuss the naming variables.

Unit-II

3. (a) Define array and explain about types of arrays in C++ with suitable example
(b) Write a C++ program using switch case. 10

(3)

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Or

Define the following control statements with example :

- (a) if-else
- (b) while
- (c) do-while

Unit-III

4. Differentiate call by value and call by reference with suitable example. 10

Or

Write notes on the following :

- (a) Pointer with example
- (b) Structure and union with example

Unit-IV

5. What is Constructor ? How many types of constructors are there ? Explain with example. 10

Or

What is polymorphism in C++ ? Explain about its types with example.

P.T.O.

Unit-V

6. Discuss about the dynamic memory allocation operators new and delete with example. Also describe virtual function. 10

Or

What is operator overloading? Explain with example. Also explain the benefits of operator overloading.